

One Number in the Weekly Report

Pick it, defend it, and name what it hides

Duration: 45 to 60 minutes

Group size: 3

Output: One Reporting Sheet per group plus a one sentence report out

Module context: Model Evaluation Metrics and Experiment Tracking. You have the confusion matrix, the four core metrics, and the metrics that survive imbalance. You do not yet have confidence intervals, cost models, or baselines beyond the majority class. Everything here is answerable with what you have.

READ THIS FIRST There is no single right answer. Every number on the candidate list is defensible and every one of them hides something specific. The work is not picking a favorite. It is being able to say, out loud, what your choice will cost the people who read it and what will be invisible to them while it looks healthy.

1. How this session runs

You will work in a group of three. Three is deliberate. Nobody gets to be a passenger, every position must be held by someone, and a group of three cannot hide a disagreement behind a majority. In your roles, you will walk through and respond to each round (as described below).

THERE ARE HINTS AND USING THEM IS NOT CHEATING Section 13 has a stuck ladder for the five hardest questions, three levels each. Take one level at a time and stop as soon as you are moving again. Reading all three levels for a question you were about to solve is the only way to waste them.

2. Your role

Each person takes one role. Roles are not decoration. Each one owns a different definition of the word good, and the friction between those definitions is the discussion. Argue your role even when you privately agree with someone else.

Role	You own	The question you keep asking
Safety team lead	The reports that go unread, and the consequences when one of them mattered	What does this number do on the week we miss a real one?
Triage operations manager	Three reviewers with finite hours, absorbing every false alarm the model produces	What does this number do to my queue, and can my team survive it going up?
Model owner and report author	The model, the threshold, and the person who has to produce this number every Monday	What is this computed on, and what will I say when it moves and I do not know why?

3. The situation

Cordwell's escalation model has been in production for five weeks. It reads incoming product reviews and flags the ones that look like genuine safety reports so the product safety team sees them within 24 hours. A miss means a safety report sits unread. A false alarm means a human reviewer spends time on a routine complaint.

It was released against a held-out test set of 2,000 reviews containing 67 genuine safety escalations, a base rate of 3.35 percent. The test set was opened once, at release, exactly as the discipline requires. Nobody has opened it since.

This morning, Product asked for something reasonable. The weekly leadership review gives every team one slide. They want one number on it, tracked week over week, so leadership can see whether the escalation model is working.

YOUR ASSIGNMENT Pick the number. Defend it. Name what it hides. Your group reports back in one sentence.

The sentence has a fixed shape, and you will find it in Section 11. Read it now, because knowing the shape of the answer changes how you argue toward it.

4. The evaluation contract

Four questions that sit underneath any metric anyone reports. They are here because at least one of them has no answer at Cordwell right now and finding out which is question 1.5.

#	Question	Why it decides everything downstream
1	What counts as a positive?	It needs a written labeling guide, not a shared instinct. If the definition drifts, every number built on it drifts with it and nothing in the pipeline notices.
2	Who labeled it, and how?	Label quality is the ceiling on model quality. You cannot interpret any metric without knowing how much two humans agree with each other.
3	How many labels do you have, and how many are positive?	The positive count, not the row count, governs how much you can conclude.
4	What does a mistake cost in each direction?	The asymmetry between a miss and a false alarm drives metric choice, threshold choice, and what an acceptable operating point even is.

5. The candidate numbers

Every metric you have been shown today, with its value on the Cordwell test set at the current threshold of 0.50. Reference card, not an answer. The last two columns are the ones that will earn their keep in Round 2.

Metric	Cordwell at 0.50	A model that always says routine	Which cells of the matrix it uses	Moves if only the threshold changes?
Accuracy	0.9395	0.9665	All four	Yes
Precision	0.3312	0.000 (genuinely undefined, filled in as zero)	TP and FP	Yes
Recall	0.7910	0.0000	TP and FN	Yes
F1	0.4670	0.0000	TP, FP, FN. Ignores TN entirely.	Yes
F2 (recall weighted twice)	0.6192	0.0000	TP, FP, FN	Yes
Balanced accuracy	0.8678	0.5000	All four	Yes
MCC	0.4880	0.0000	All four	Yes
ROC-AUC	0.9621	0.5000	Derived from scores across every threshold	No
Average precision	0.6040	0.0335 (the base rate)	Derived from scores across every threshold	No

The confusion matrix those come from

Test set: 2,000 reviews, 67 genuine safety escalations, threshold 0.50

	Predicted: Escalate	Predicted: Routine
Actual: Safety	TP = 53	FN = 14
Actual: Routine	FP = 107	TN = 1,826

Flag rate: 160 of 2,000 = 8.0 percent

Nothing stops you proposing a number that is not on this card. Some of the strongest answers are not classification metrics at all.

6. What production records

This is the most important section in the packet. Read it twice and read it before Round 1.

Recorded every week, automatically	Not recorded at all
How many reviews came in	Whether an unflagged review was actually a safety report
How many the model flagged	Anything at all about the reviews that were not flagged
Which of the flagged ones the triage team confirmed as genuine	The channel and product category breakdown of model performance
Which of the flagged ones the triage team rejected as routine	How often the two safety-team labelers would disagree with each other
Median time from flag to triage	Any label on any production review that the model did not flag

Item	Detail
Labeling of the test set	Two members of the safety team, working from a written guide. Their agreement with each other has never been measured.
Production labels	None, beyond the triage team's confirm or reject decision on the reviews the model flagged.
Threshold	0.50. Changing it is a one-line config change that any engineer on the team can make and ship the same day.
Channels	Reviews arrive through the web store, the mobile app, and in-store kiosks. No metric is reported per channel.
Test set	Opened once at release. The discipline says it is a one-shot instrument, one look per release.
Triage team	Three reviewers. Comfortable capacity is about 120 flagged reviews per week.
Team	Two engineers on the model. Neither has spare time this quarter.

7. What a week looks like

Carried over from the test set numbers onto the production volume. Check the arithmetic if you want to; it is all from Section 5.

Quantity	Per week
Reviews received	About 1,200
Reviews the model flags, at an 8 percent flag rate	About 96
Genuine safety reports among them, at the 3.35 percent base rate	About 40

Quantity	Per week
Of those 40, how many the model catches at recall 0.791	About 32
Of those 40, how many go unread	About 8
Reviews nobody looks at, because the model did not flag them	About 1,104
Triage capacity	About 120

WORTH SITTING WITH About eight genuine safety reports go unread every week. Over a year that is more than four hundred. None of them appear in any record Cordwell keeps, because nobody looks at the 1,104 reviews the model did not flag.

8. Round 0: your first pick

Silent. Individual. Do not discuss.

Write four things, then stop.

- The number you would put on the slide.
- The one thing it hides.
- The one question you would ask Product before agreeing to any of this.
- Your confidence, one to five.

The third line is the one that matters. Groups where somebody wrote a sharp question there tend to end the session with a much better answer than groups where everyone went straight to a metric name.

9. Round 1: what is the report actually for

Group. Do not choose a metric yet.

The rule this round: nobody names their pick. If someone starts defending a metric, then stop. A number is only good relative to a decision, and you have not established what the decisions are.

- 1.1** Who reads this weekly report, and what does each reader do differently on a bad week? Name at least three readers, and for each one name the decision the number is supposed to inform. Then answer the question that follows from your own list: can one number serve all of them?

- 1.2 Look at Section 6. Given only what production records, which of the metrics on the Section 5 card can you actually compute every week, and which cannot? Work through the card row by row. The answer is shorter than you expect.
- 1.3 For the metrics you cannot compute weekly, say what it would take to make them computable. Who does what work, roughly how much of it, and how often. Section 7 has the numbers you need to size it.
- 1.4 The test set is a one-shot instrument, opened once per release. A weekly number implies weekly recomputation. Reconcile those two. There are only a few possible sources for a weekly number, and you can name all of them. Say which one you would use and what you are giving up.

10. Round 2: pick it, defend it, name what it hides

Group. Fill the Reporting Sheet in Section 12 as you go.

- 2.1 Pick your number. Write the defense in this shape: we report [metric], because it drives [decision] for [reader]. A defense that does not name a reader and a decision is a preference, not a defense.
- 2.2 Name what it hides. Not generically. Name a specific failure that could be happening this week, right now, while your number looks perfectly healthy on the slide.
- 2.3 The gauntlet. Four people read your number. For each one, say what they do when it looks good and what they do when it drops. Then say which of the four is worst served by your choice, and whether that is acceptable.

Reader	What they are trying to decide
VP of Customer Experience	Whether this model is worth continuing to fund next year
The safety team lead	Whether anyone is getting hurt by what the system misses
The triage operations manager	Whether she needs to ask for a fourth reviewer
The engineer who inherits this next quarter	What changed, when something moves and nobody remembers why

- 2.4 Next quarter your number is in the OKRs with a numeric target. Name three ways somebody could move it without the model getting any better at finding safety reports. At least one of your three must involve neither the model nor the threshold.
- 2.5 Name the number you would refuse to put on that slide and say why reporting it would be worse than reporting nothing at all. There is at least one obvious candidate on the card and at least one less obvious one.
- 2.6 Product relents and gives you a second number, free. Which one, and what does the pair let you say that neither number says alone? A good pair fails in opposite directions.

11. Round 4: report out

The sentence has a fixed shape. Fill in the four blanks and read it exactly as written. The constraint is the point: if you cannot fit your defense into this sentence, you have not finished deciding.

We report _____, computed on _____, because it drives _____ for _____, and it hides _____.

Then three additions, one short phrase each.

#	Add this	What makes it count
1	Your second number	And in one clause, what the pair says that neither says alone.
2	The number you refuse to report	With the reason. "It is misleading" is not a reason. Say what it would cause someone to do.
3	Where your group disagreed	Not optional. In a group of three there is no majority, so a clean consensus means somebody gave in.

Note the second blank in the sentence. It is there because most groups get most of the way through this exercise before anyone asks what the number is computed on, and it is the hardest part of the whole problem.

12. Your deliverable: the Reporting Sheet

One per group. Hand it to your facilitator at the end.

Part 1. The sentence

Part 2. The gauntlet

Reader	What they do when it looks good	What they do when it drops	Well served?
VP of Customer Experience			
Safety team lead			
Triage operations manager			
The next engineer			

Part 3. Three ways this number could move without the model improving

13. If your group is stuck

Three levels per question. Take one at a time and stop as soon as you are moving again. None of these contains an answer and reading all three when level one would have done is the only way to waste them.

Question 1.1 | Readers and decisions

Level 1. A number is only useful relative to a decision. Start from the other end. What decisions actually get made about this system in a given quarter, and by whom?

Level 2. Section 2 puts two of those people in the room with you, and Section 10 names four readers. Product is not any of them. For each, ask what they would physically do differently on a week when the number looks bad.

Level 3. One reader is deciding whether to keep funding this. One is deciding whether anybody got hurt. One is deciding whether to ask for another headcount. Those are three different questions, and only one of them is really about model quality. Now say whether a single number can answer all three, and what it would have to be if it could.

Question 1.2 | What you can compute weekly

Level 1. Take the Section 5 card one row at a time. Each metric needs particular cells of the confusion matrix. Ask, for each, whether production hands you those cells.

Level 2. The triage team confirms or rejects everything the model flags. Nobody looks at anything the model does not flag. Write down which of TP, FP, FN, and TN that gives you, and which it leaves empty.

Level 3. If you never inspect the unflagged pile, you know how many of your flags were right. You do not know how many flags you failed to make. Go back to the card and cross out every metric whose formula needs the cell you do not have. Notice how few rows survive and notice which single quantity almost all of them depend on.

Question 1.4 | Weekly, from what

Level 1. "Weekly" and "once per release" are in direct tension. Say the tension out loud before anyone tries to resolve it.

Level 2. There are only three plausible sources for a weekly number, and your group can name all three without help. One of them is the test set, and Section 6 tells you what using it weekly would destroy.

Level 3. Fresh labels every week cost human hours. Production signals are free but incomplete, for the reason you found in 1.2. Reusing the test set is free and quietly ruins the only instrument you have for gating a release. There is no option without a cost, so the answer is which cost you are choosing and saying so on the slide.

Question 2.4 | Moving the number without improving the model

Level 1. Look at the last column of the Section 5 card. Anything answering yes there can be moved by one person with config access in an afternoon.

Level 2. That is one way and it is the easy one. Now find a way that changes nothing about the model or the threshold and instead changes the population the number is computed over.

Level 3. A metric is a ratio over a population, using a definition of what counts. So there are three surfaces to push on: the numerator, the denominator, and the definition. Section 6 tells you who decides what counts as a confirmed safety issue, and Section 4 question one tells you that definition is a document, written by people, that can be rewritten.

Question 2.6 | The second number

Level 1. A pair earns its place when the two numbers fail in opposite directions. Ask what would have to be true for one person to move both of them at once.

Level 2. Most of the candidates on the card move together when the threshold moves, so most pairs are not really a pair. Look for a second number that moves the other way, or that does not move at all when the threshold does.

Level 3. Two pairings tend to work. One is a quality number next to a capacity number, because the capacity number is precisely the constraint the quality number ignores. The other is a number next to its own baseline, because the third column of the Section 5 card already tells you what a useless model scores on every candidate.

14. Vocabulary for this discussion

Term	Working definition in this context
Base rate	The fraction of examples that are genuinely positive. 3.35 percent here. It is the honest starting line for average precision, and it is why accuracy is nearly meaningless on this problem.
Majority baseline	A model that always predicts the dominant class. It scores 0.9665 accuracy at Cordwell and finds zero safety reports. Any metric it can score well on is a metric you should not report.
Threshold dependent	A metric that changes when you change the cut-off, with no change to the model. Most of the card is threshold dependent, which matters enormously once a number becomes a target.
Aggregate	A number computed over the whole population. It can look healthy while one channel or one product category fails completely.
Ceiling	The best score any model could achieve, set by how much the human labelers agree with each other. It is not 1.0, and nobody at Cordwell has measured it.
Offline and online evaluation	Offline runs against a held-out labeled test set before release. Online measures real behavior in production. They answer different questions and neither substitutes for the other.
Goodhart's law	When a measure becomes a target, it stops being a good measure. The practical version: name the three cheapest ways to hit the target without doing the work and assume somebody eventually will.